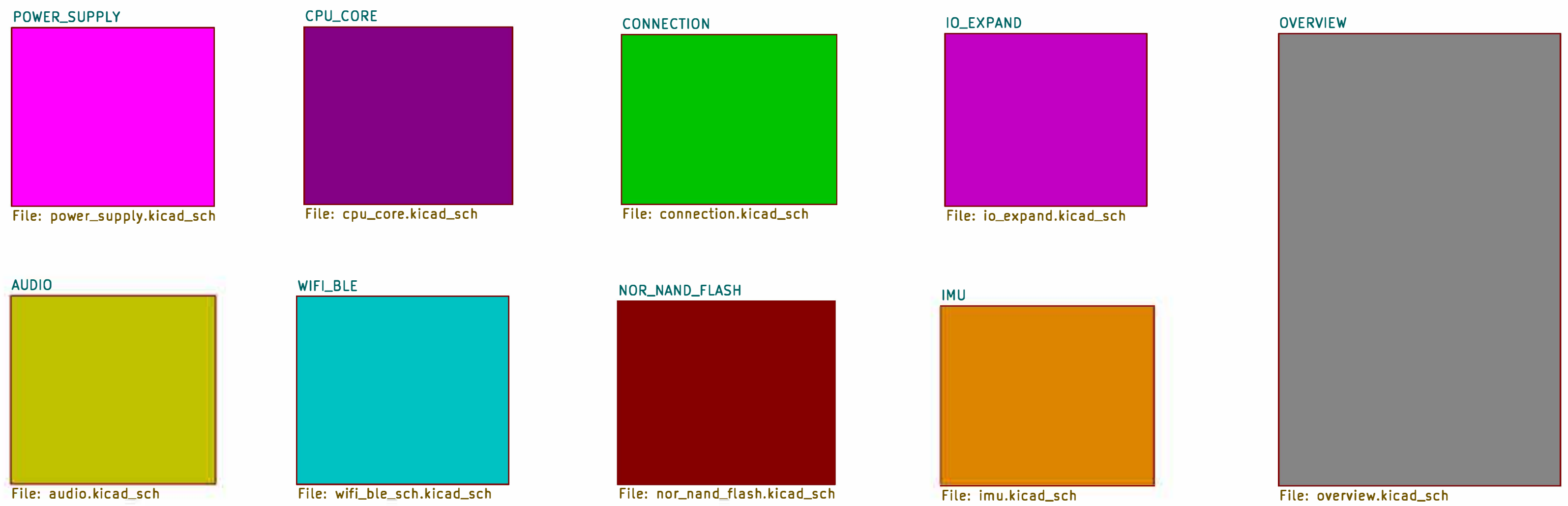
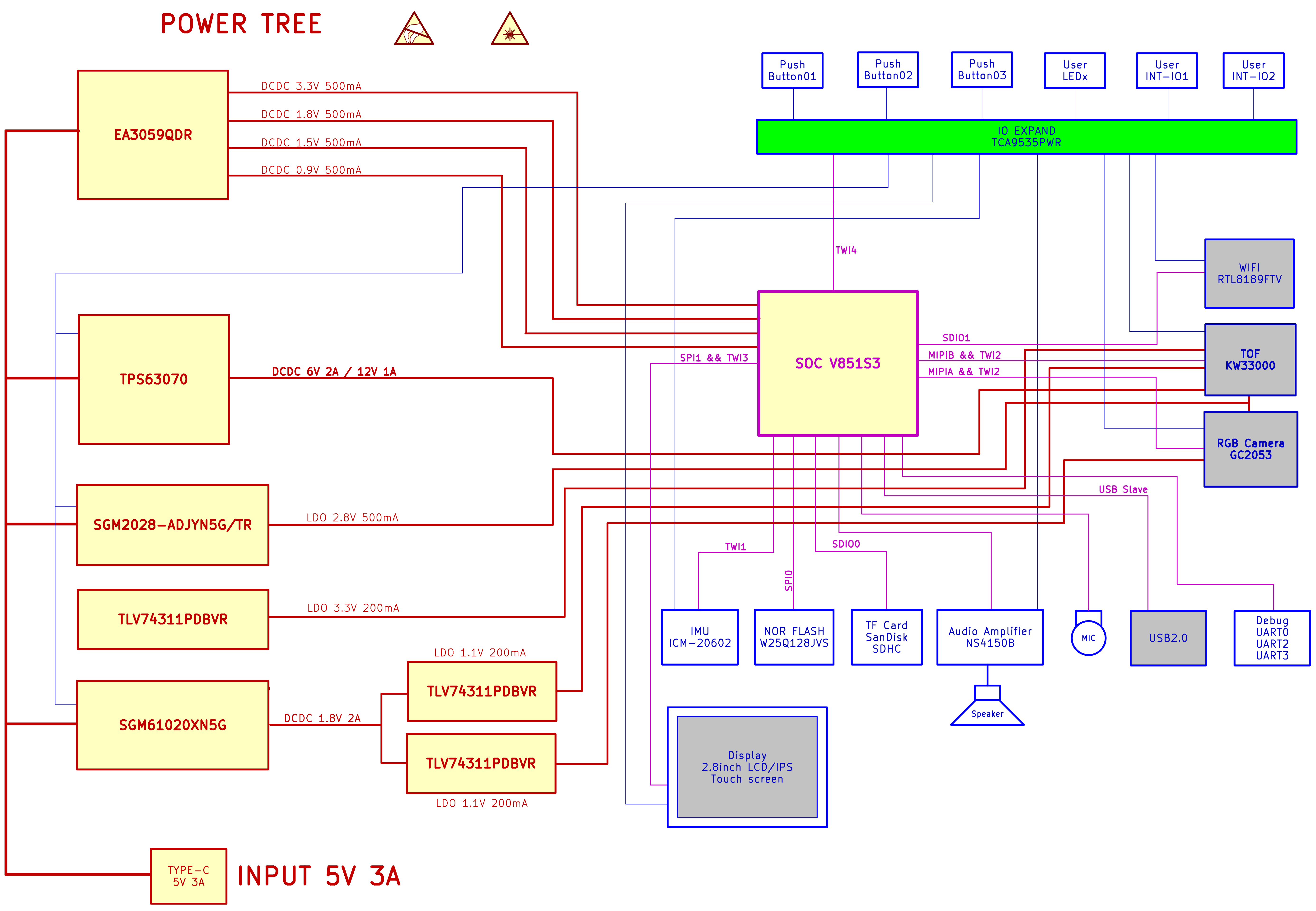
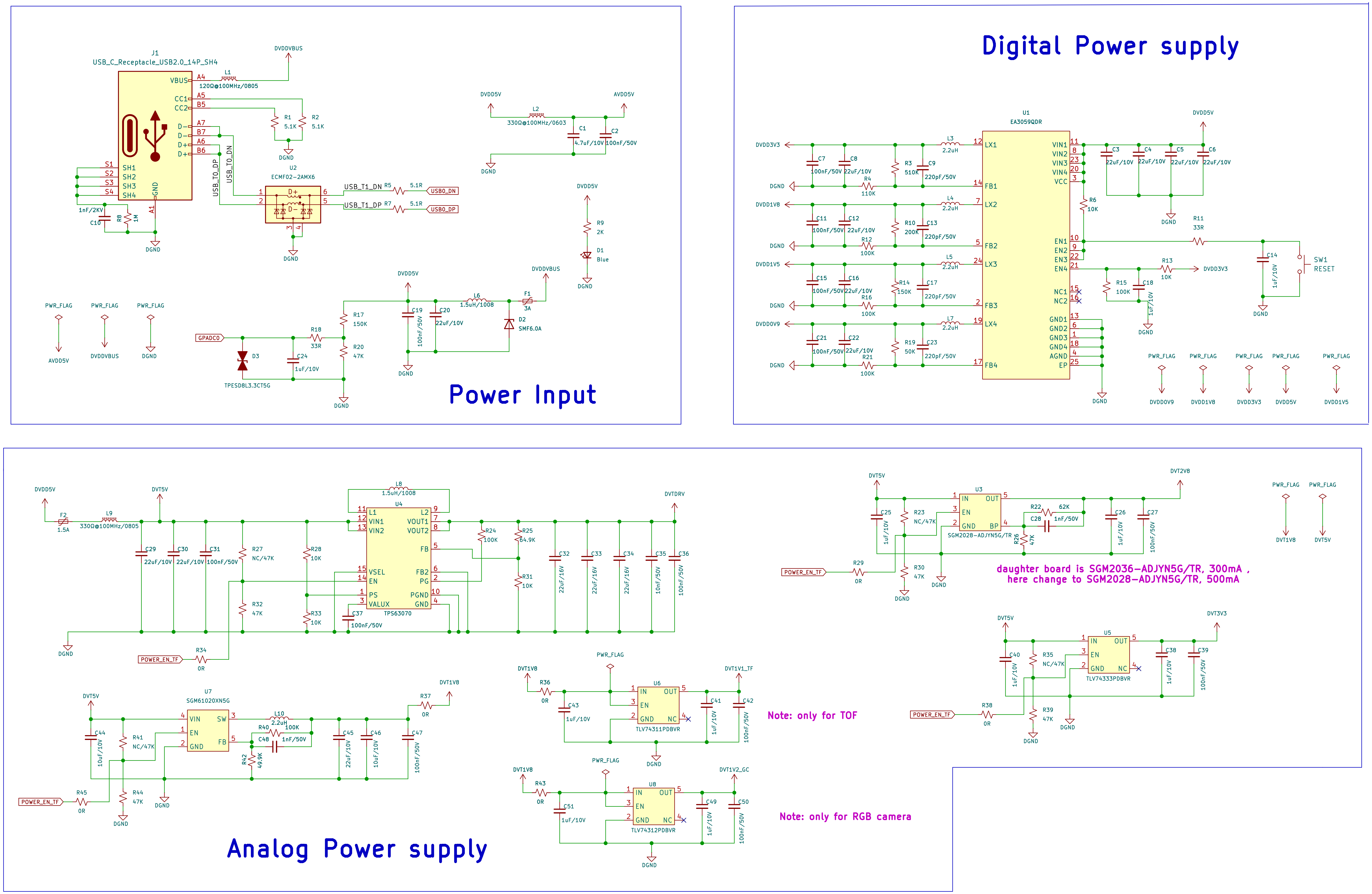
NEW TOF and RGB Stereo AI Camera

Ver	Date	Editor	Note
V0.1	2025.12.29		finish the TOF camera demo board
V0.11	2025.12.30		modify the TOF 5V power part, changed to 3.3V
V0.11	2026.01.04		1. add a BLUE led to the VLD_VCC 2. modify the User led drive mode, change to low voltage output from the TCA9535PWR, refer to the datasheet 3. add the 2.2uF and 100nF capacity to the VCC_LDO (pin 14) of SOC even not use coz the PLL and internal peripheral maybe use
V0.11	2026.01.05		1. add RESET push button in EA3059QDR, the SOC use its internal reset function, here reset the total power tree to reset the SOC 2. modify the BOOT update FEL pin from SPI0_MOSI to SPI0_CS 3. add 10nF capacity to the FEL push button
V0.11	2026.01.06		add 0R resistor to Pin 79 of SOC for debug

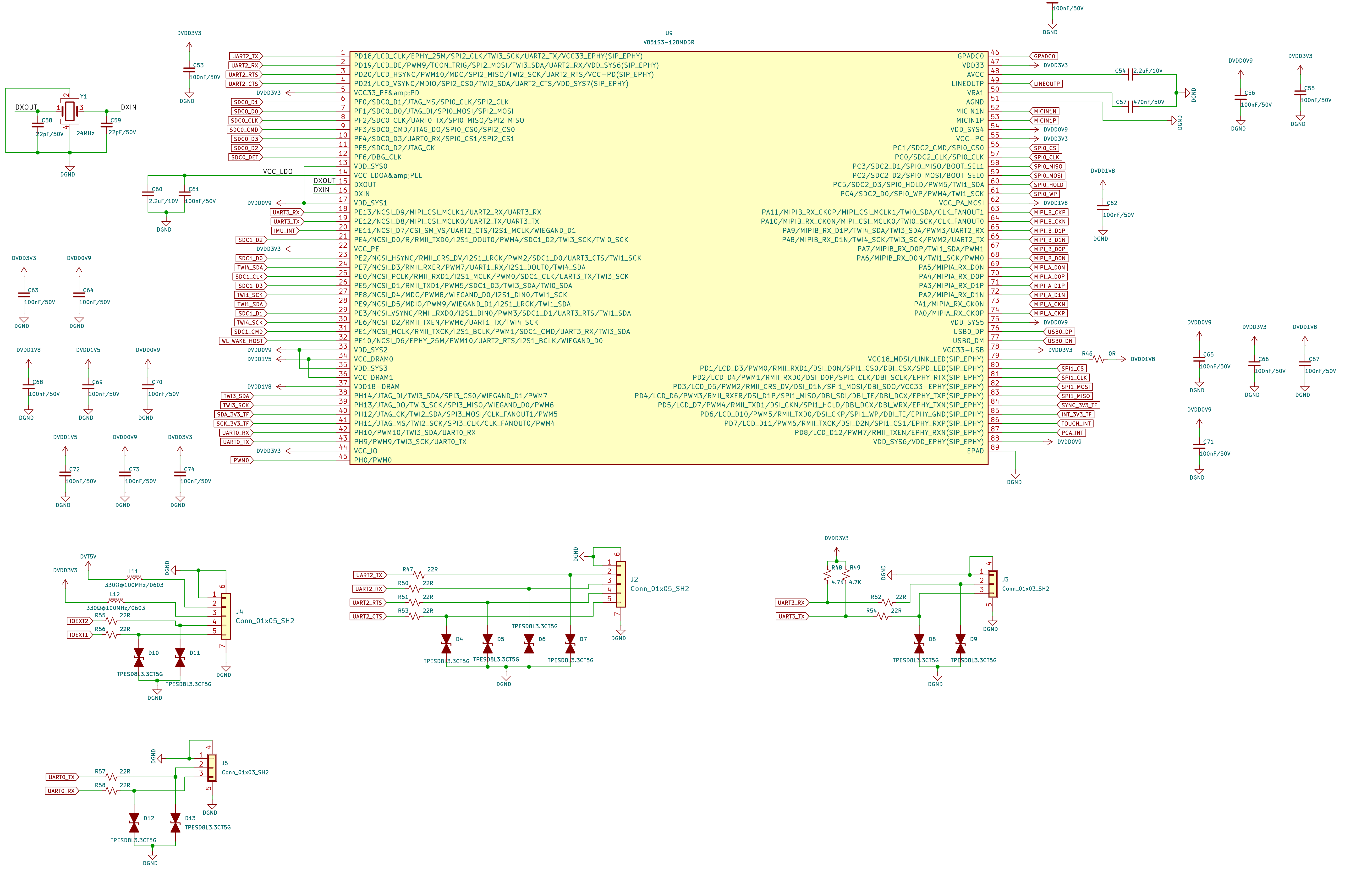


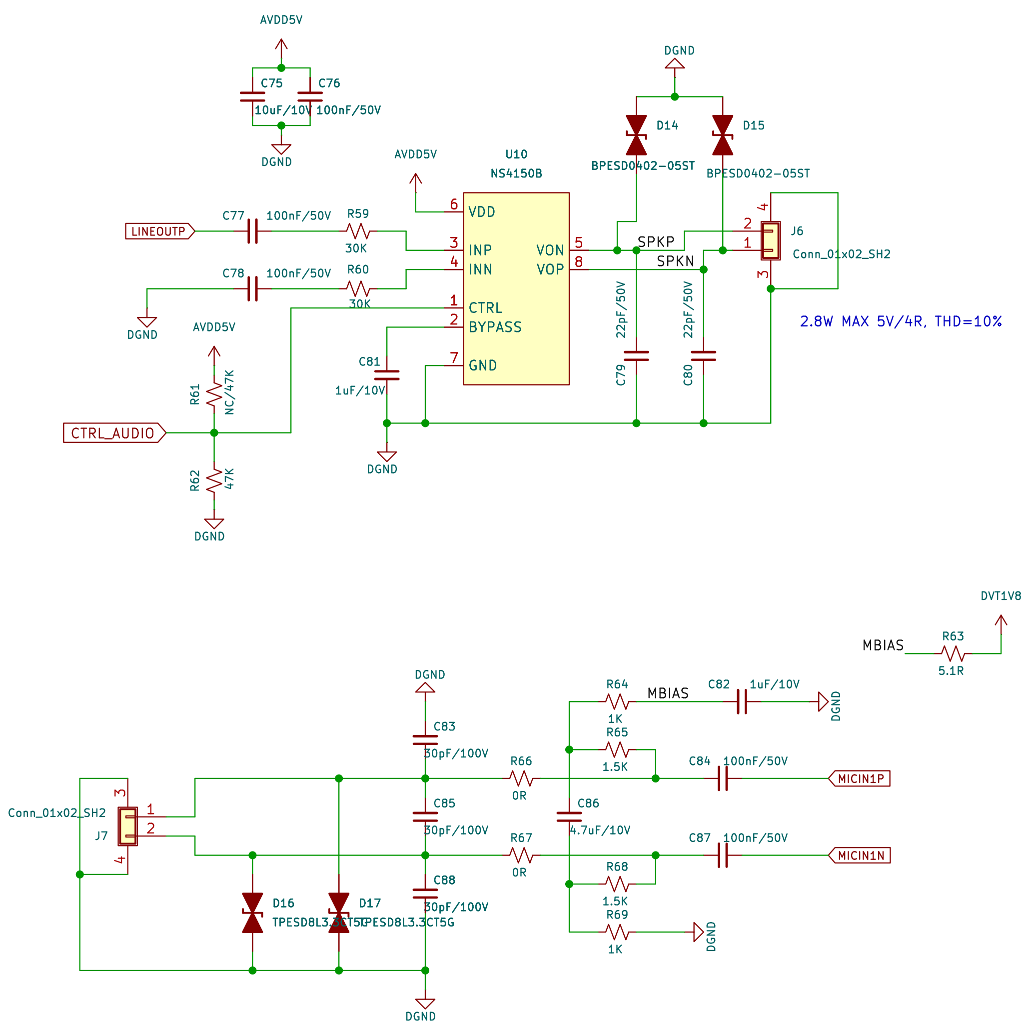


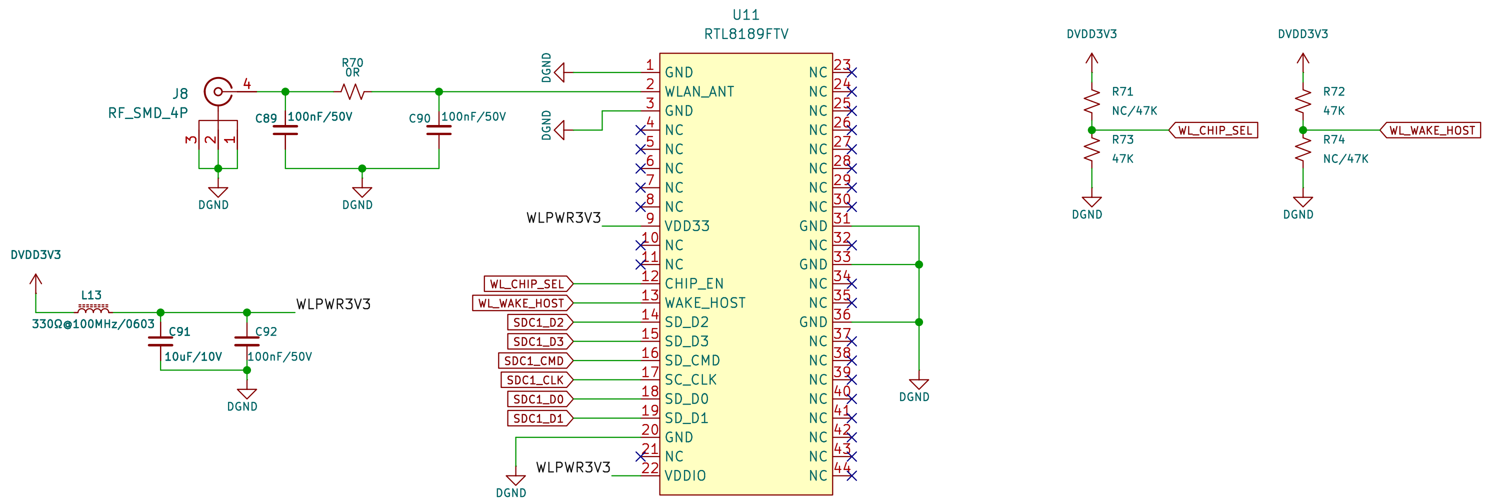
POWER TREE
EA3059QDR
DCDC 3.3V 500mA
DCDC 1.8V 500mA
DCDC 1.5V 500mA
DCDC 0.9V 500mA
TPS63070
DCDC 6V 2A / 12V 1A
SGM2028-ADJYN5G/TR
LDO 2.8V 500mA
TLV74311PDBVR
LDO 3.3V 200mA
SGM61020XN5G
DCDC 1.8V 2A
LDO 1.1V 200mA
TLV74311PDBVR
TLV74311PDBVR
LDO 1.1V 200mA
TYPE-C
5V 3A
INPUT 5V 3A
Push
Button01
Push
Button02
Push
Button03
User
LEDx
User
INT-IO1
User
INT-IO2
IO EXPAND
TCA9535PWR
TWI4
WIFI
RTL8189FTV
SOC V851S3
SPI1 && TWI3
SDIO1
MIPIB && TWI2
MIPIA && TWI2
TOF
KW33000
RGB Camera
GC2053
USB Slave
TWI1
SPI0
SDIO0
IMU
ICM-20602
NOR FLASH
W25Q128JVS
TF Card
SanDisk
SDHC
Audio Amplifier
NS4150B
MIC
USB2.0
Debug
UART0
UART2
UART3
Speaker
Display
2.8inch LCD/IPS
Touch screen


Power Input
Digital Power supply
Analog Power supply
J1
USB_C_Receptacle_USB2.0_14P_SH4
DVDDVBUS
120Ω@100MHz/0805
U2
ECMF02-2AMX6
USB_T1_DN
USB_T1_DP
USB0_DN
USB0_DP
DVDD5V
AVDD5V
330Ω@100MHz/0603
4.7uF/10V
100nF/50V
DGND
GPADC0
TPESD8L3.3CT5G
1.5uH/1008
SMF6.0A
3A
Blue
PWR_FLAG
U1
EA3059QDR
DVDD3V3
DVDD1V8
DVDD1V5
DVDD0V9
SW1
RESET
U4
TPS63070
DVTDRV
DVT5V
POWER_EN_TF
U3
SGM2028-ADJYN5G/TR
DVT2V8
daughter board is SGM2036-ADJYN5G/TR, 300mA ,
here change to SGM2028-ADJYN5G/TR, 500mA
U5
TLV74333PDBVR
DVT3V3
U7
SGM61020XN5G
DVT1V8
U6
TLV74311PDBVR
DVT1V1_TF
Note: only for TOF
U8
TLV74312PDBVR
DVT1V2_GC
Note: only for RGB camera

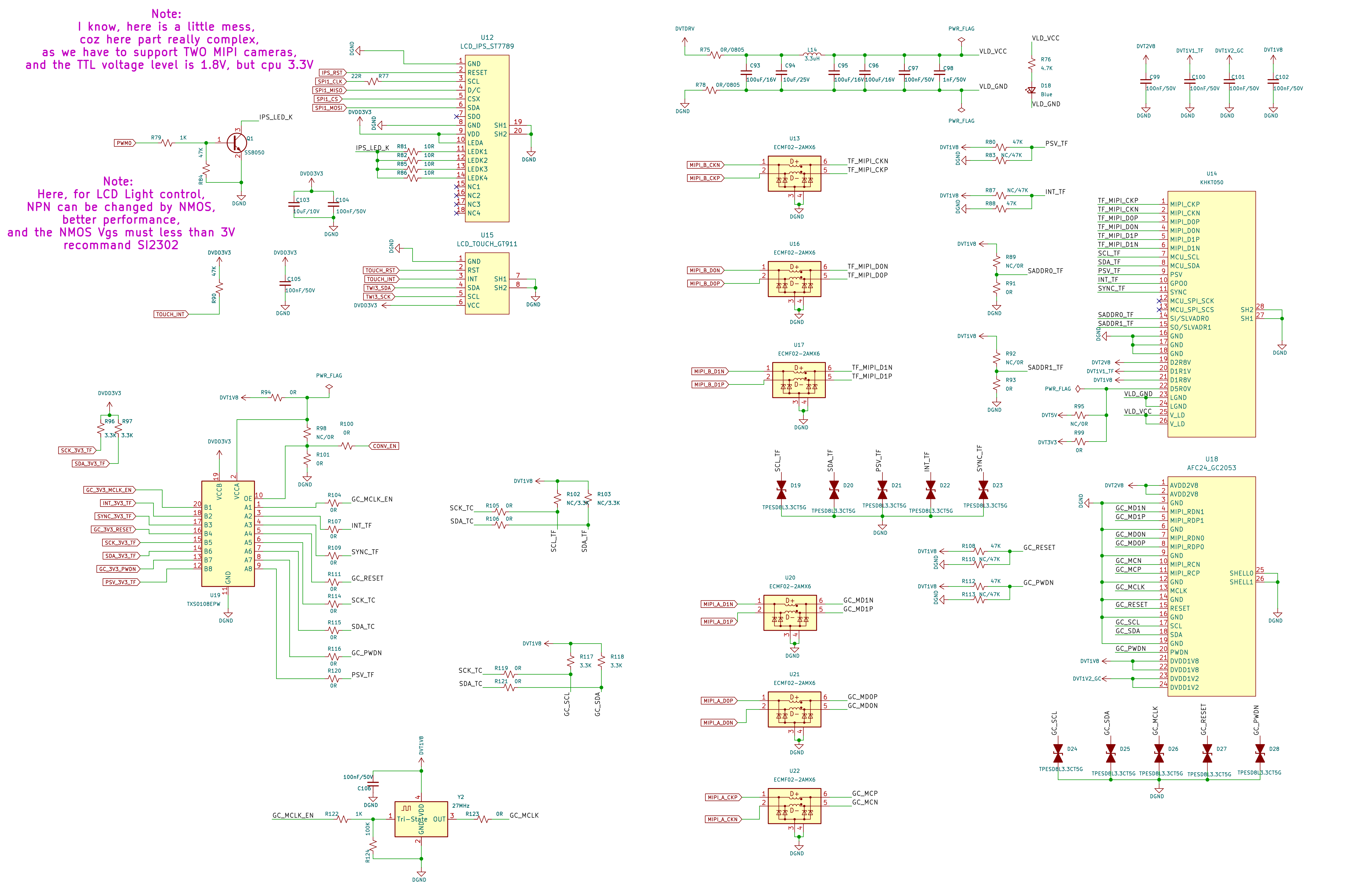
Note:
LOOK, use all of the soc PINS,
NO any waste... this is NOT gready, okay
that's not easy for the design...

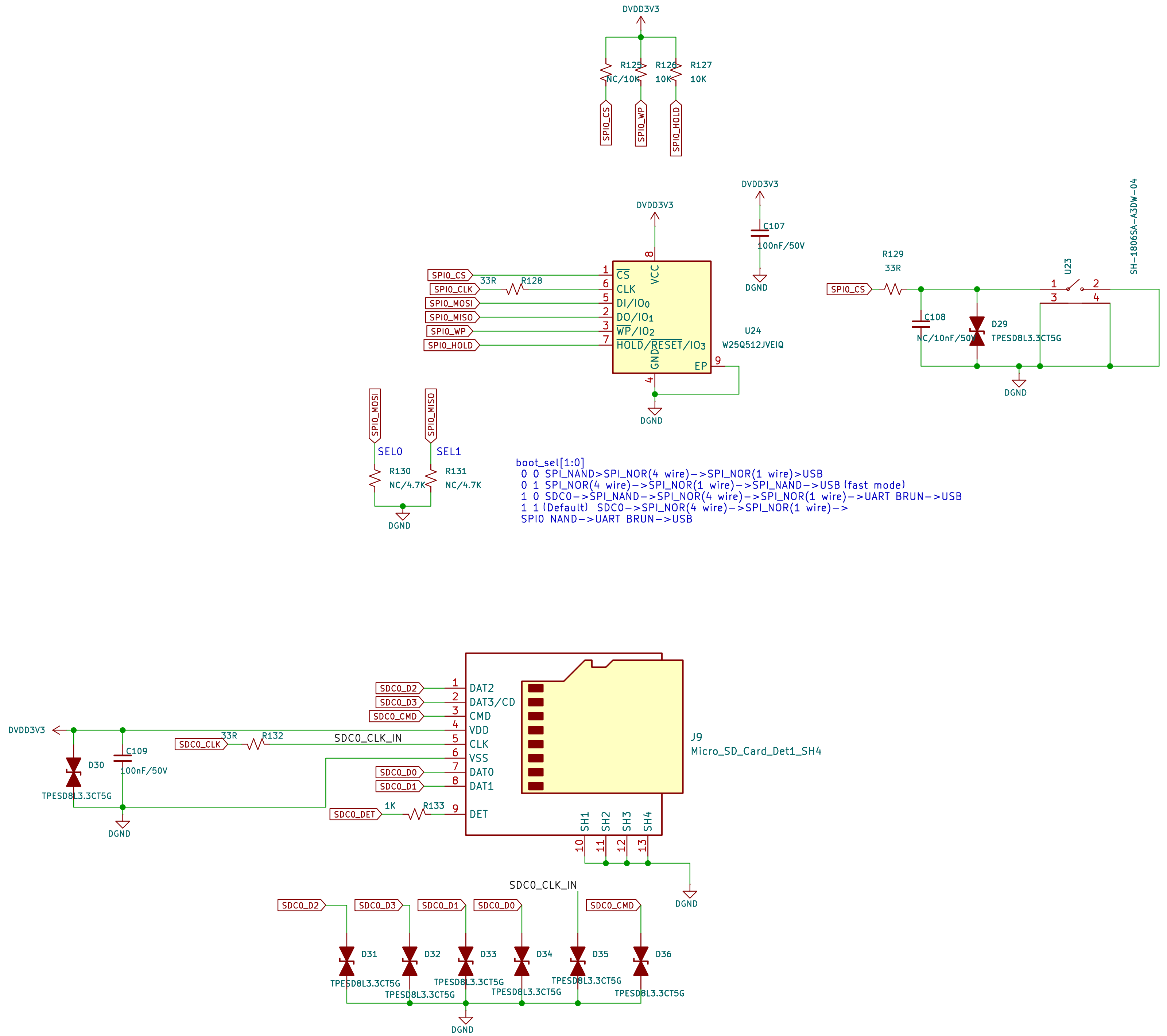



AVDD5V
C75
10uF/10V
C76
100nF/50V
DGND
DGND
D14
BPESD0402-05ST
D15
BPESD0402-05ST
AVDD5V
U10
NS4150B
VDD
INP
INN
CTRL
BYPASS
GND
VON
VOP
SPKP
SPKN
J6
Conn_01x02_SH2
LINEOUTP
C77
100nF/50V
R59
30K
C78
100nF/50V
R60
30K
C81
1uF/10V
22pF/50V
C79
22pF/50V
C80
2.8W MAX 5V/4R, THD=10%
AVDD5V
R61
NC/47K
CTRL_AUDIO
R62
47K
DGND
DVT1V8
MBIAS
R63
5.1R
DGND
C83
30pF/100V
C85
30pF/100V
C88
30pF/100V
R64
1K
MBIAS
C82
1uF/10V
R65
1.5K
C84
100nF/50V
MICIN1P
R66
0R
C86
4.7uF/10V
R67
0R
C87
100nF/50V
MICIN1N
R68
1.5K
R69
1K
Conn_01x02_SH2
J7
D16
TPESDBL3.3CT5G
D17
TPESDBL3.3CT5G
DGND

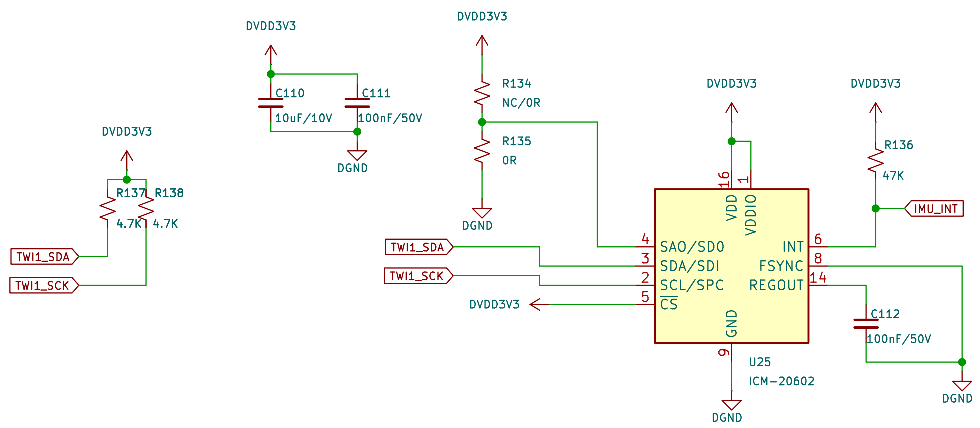


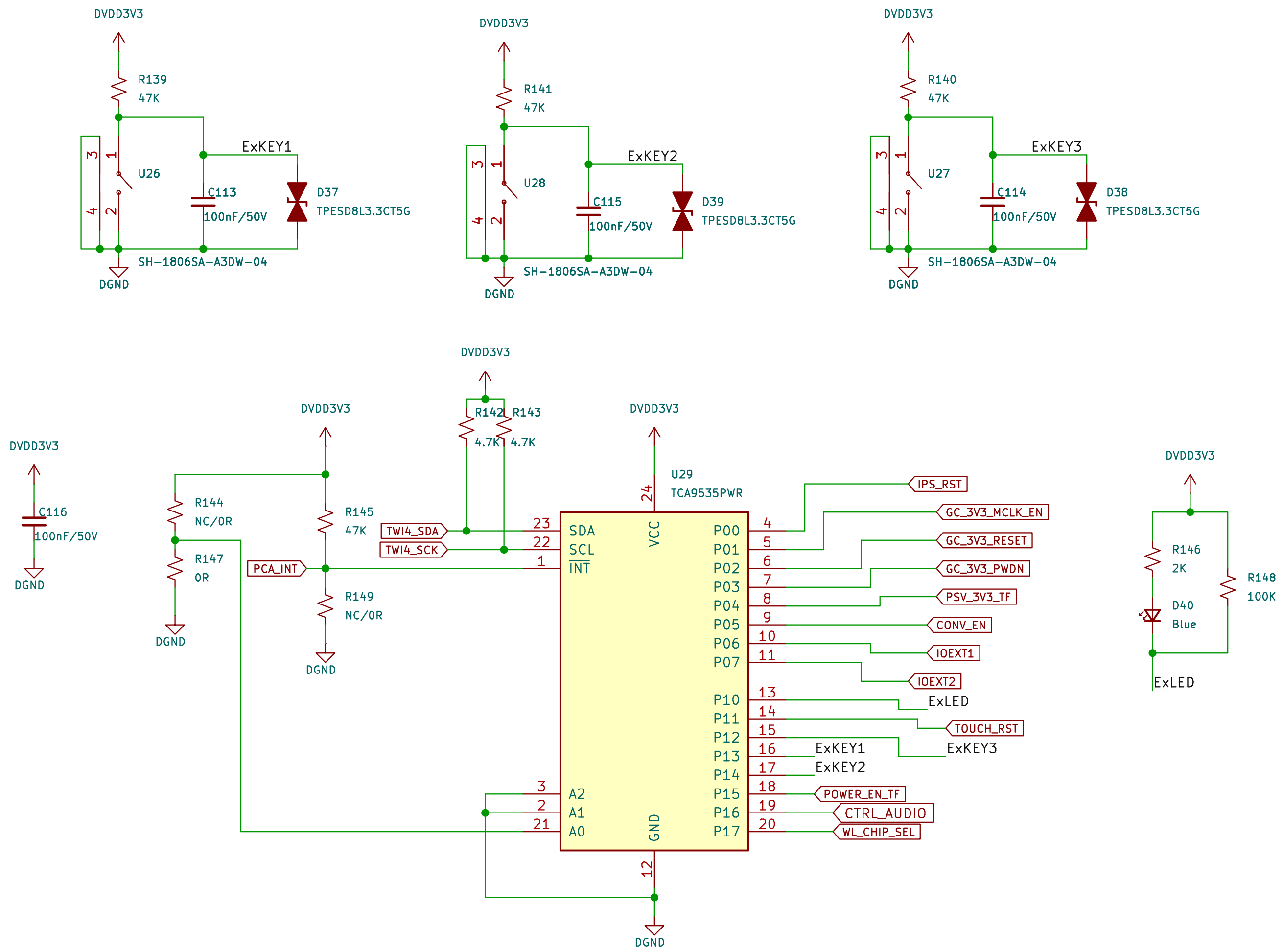
U11
RTL8189FTV
J8
RF_SMD_4P
R70
0R
C89
100nF/50V
C90
100nF/50V
DGND
L13
330Ω@100MHz/0603
C91
10uF/10V
C92
100nF/50V
DVDD3V3
WLPWR3V3
GND
WLAN_ANT
NC
VDD33
CHIP_EN
WAKE_HOST
SD_D2
SD_D3
SD_CMD
SC_CLK
SD_D0
SD_D1
VDDIO
WL_CHIP_SEL
WL_WAKE_HOST
SDC1_D2
SDC1_D3
SDC1_CMD
SDC1_CLK
SDC1_D0
SDC1_D1
R71
NC/47K
R73
47K
R72
47K
R74
NC/47K


Note:
I know, here is a little mess,
coz here part really complex,
as we have to support TWO MIPI cameras,
and the TTL voltage level is 1.8V, but cpu 3.3V
Note:
Here, for LCD Light control,
NPN can be changed by NMOS,
better performance,
and the NMOS Vgs must less than 3V
recommand SI2302
U12
LCD_IPS_ST7789
U15
LCD_TOUCH_GT911
U19
TXS0108EPW
Y2
27MHz
Tri-State
U13
ECMF02-2AMX6
U16
ECMF02-2AMX6
U17
ECMF02-2AMX6
U20
ECMF02-2AMX6
U21
ECMF02-2AMX6
U22
ECMF02-2AMX6
U14
KHKT050
U18
AFC24_GC2053
L14
3.3uH
D18
Blue
Q1
SS8050
IPS_LED_K
PWM0
TOUCH_INT
VLD_VCC
VLD_GND
PWR_FLAG
DVTDRV
GC_MCLK_EN
GC_MCLK
TPESD8L3.3CT5G



DVDD3V3
R125
NC/10K
R126
10K
R127
10K
SPI0_CS
SPI0_WP
SPI0_HOLD
DVDD3V3
C107
100nF/50V
DGND
DVDD3V3
SPI0_CS
SPI0_CLK
SPI0_MOSI
SPI0_MISO
SPI0_WP
SPI0_HOLD
33R
R128
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
EP
U24
W25Q512JVEIQ
DGND
R129
33R
SPI0_CS
C108
NC/10nF/50V
D29
TPESD8L3.3CT5G
U23
SH-1806SA-A3DW-04
DGND
SPI0_MOSI
SPI0_MISO
SEL0
SEL1
R130
NC/4.7K
R131
NC/4.7K
DGND
boot_sel[1:0]
0 0 SPI_NAND>SPI_NOR(4 wire)->SPI_NOR(1 wire)>USB
0 1 SPI_NOR(4 wire)->SPI_NOR(1 wire)->SPI_NAND->USB (fast mode)
1 0 SDC0->SPI_NAND->SPI_NOR(4 wire)->SPI_NOR(1 wire)->UART BRUN->USB
1 1 (Default) SDC0->SPI_NOR(4 wire)->SPI_NOR(1 wire)->
SPI0 NAND->UART BRUN->USB
DVDD3V3
D30
TPESD8L3.3CT5G
C109
100nF/50V
DGND
SDC0_CLK
33R
R132
SDC0_CLK_IN
SDC0_D2
SDC0_D3
SDC0_CMD
SDC0_D0
SDC0_D1
SDC0_DET
1K
R133
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SH1
SH2
SH3
SH4
J9
Micro_SD_Card_Det1_SH4
DGND
SDC0_CLK_IN
SDC0_D2
SDC0_D3
SDC0_D1
SDC0_D0
SDC0_CMD
D31
D32
D33
D34
D35
D36
TPESD8L3.3CT5G
TPESD8L3.3CT5G
TPESD8L3.3CT5G
TPESD8L3.3CT5G
TPESD8L3.3CT5G
TPESD8L3.3CT5G
DGND


DVDD3V3
R137
R138
4.7K
4.7K
TWI1_SDA
TWI1_SCK
DVDD3V3
C110
10uF/10V
C111
100nF/50V
DGND
DVDD3V3
R134
NC/0R
R135
0R
DGND
TWI1_SDA
TWI1_SCK
DVDD3V3
DVDD3V3
VDD
VDDIO
SAO/SDO
SDA/SDI
SCL/SPC
CS
INT
FSYNC
REGOUT
GND
U25
ICM-20602
DGND
DVDD3V3
R136
47K
IMU_INT
C112
100nF/50V
DGND



DVDD3V3
R139
47K
U26
C113
100nF/50V
ExKEY1
D37
TPESD8L3.3CT5G
SH-1806SA-A3DW-04
DGND
R141
47K
U28
C115
100nF/50V
ExKEY2
D39
TPESD8L3.3CT5G
R140
47K
U27
C114
ExKEY3
D38
R142
R143
4.7K
C116
R144
NC/0R
R145
47K
R147
0R
R149
NC/0R
TWI4_SDA
TWI4_SCK
PCA_INT
U29
TCA9535PWR
SDA
SCL
INT
A2
A1
A0
VCC
GND
P00
P01
P02
P03
P04
P05
P06
P07
P10
P11
P12
P13
P14
P15
P16
P17
IPS_RST
GC_3V3_MCLK_EN
GC_3V3_RESET
GC_3V3_PWDN
PSV_3V3_TF
CONV_EN
IOEXT1
IOEXT2
ExLED
TOUCH_RST
POWER_EN_TF
CTRL_AUDIO
WL_CHIP_SEL
R146
2K
D40
Blue
R148
100K